DISCRETE STRUCTURES

CMSC250

Chapter 4: Arguments and Rules of Inference

By Naren Ayinala

Validity, critical rows, and step-by-step logical reasoning



Notation. We use $\sim p$ for negation (also commonly written $\neg p$). In truth tables, 1 means true and 0 means false.

Contents

1 From statements to arguments **3**

2 What validity actually means **3**

2.1 Why false premises do not automatically make an argument invalid 3

3 Critical rows in truth tables **3**

3.1 A reliable truth-table procedure 4

4 Rules of inference **4**

4.1 Modus ponens (MP) 4

4.2 Modus tollens (MT) 5

4.3 Hypothetical syllogism (HS) 5

4.4 Disjunctive syllogism (DS) 5

4.5 Conjunction, simplification, and addition 5

4.6 Resolution 5

5 Formal derivations with numbered lines **5**

6 Inference rules versus equivalence laws **6**

7 Common invalid argument forms **7**

7.1 Affirming the consequent 7

7.2 Denying the antecedent 7

7.3 Confusing $\vee$ with exclusive choice 7

8 Choosing a method **7**

9 Common mistakes and grading pitfalls **8**

10 Practice problems **8**

11 Detailed solutions **9**

1 From statements to arguments

A statement can be true or false. An **argument** does something different: it presents one or more statements as **premises** and claims that another statement, the **conclusion**, follows from them. The central question is not whether the conclusion happens to be true. It is whether the premises *force* the conclusion to be true.

For example, consider the argument: "If the alarm sounds, the building is evacuated. The alarm sounds. Therefore, the building is evacuated." Let a mean "the alarm sounds" and e mean "the building is evacuated." Its logical form is

$$\underbrace{a \to e}_{\text{premise 1}}, \qquad \underbrace{a}_{\text{premise 2}} \qquad \therefore \qquad \underbrace{e}_{\text{conclusion}}.$$

This is a **valid argument**. No interpretation makes both premises true while the conclusion is false.

Key idea. Validity concerns the relationship between premises and conclusion, not whether the premises describe the real world accurately.

2 What validity actually means

An argument with premises $P_1, P_2, \ldots, P_n$ and conclusion C is **valid** exactly when every interpretation that makes *all* premises true also makes C true. Equivalently, the formula

$$\boxed{(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \to C}$$

is a tautology.

An argument is **invalid** if there is at least one interpretation in which every premise is true and the conclusion is false. That interpretation is a **counterexample to validity**.

2.1 Why false premises do not automatically make an argument invalid

Consider "If 2 is odd, then 4 is odd. 2 is odd. Therefore, 4 is odd." The argument has the same valid *form* as the alarm argument, even though its premises are not true. Validity is about what *would* follow if the premises were true. When an argument is valid and all its premises are actually true, it is called **sound**.

3 Critical rows in truth tables

A **critical row** is a row where *every premise* has truth value 1. These are the only rows that can determine whether an argument is invalid. If no critical row has a false conclusion, the argument is valid.

Worked example: checking modus ponens with critical rows

Premises: $p \to q$ and p. Conclusion: q.

p	q	$p \to q$	p	q	Interpretation
1	1	1	1	1	Critical: conclusion true
1	0	0	1	0	Not critical
0	1	1	0	1	Not critical
0	0	1	0	0	Not critical

The first row is the only critical row. Its conclusion is 1, so the argument is valid. Notice that the last row has a false conclusion, but this does *not* make the argument invalid: premise p is false there.

Worked example: finding a counterexample to an invalid argument

Premises: $p \to q$ and q. Conclusion: p.

p	q	$p \to q$	q	p	Interpretation
1	1	1	1	1	Critical: conclusion true
1	0	0	0	1	Not critical
0	1	1	1	0	**Critical: conclusion false**
0	0	1	0	0	Not critical

The row $p = 0, q = 1$ makes both premises true and the conclusion false. Therefore the argument is invalid. This fallacy is called **affirming the consequent**.

3.1 A reliable truth-table procedure

1. List the propositional variables alphabetically. For n variables, prepare 2^n rows in descending binary order.
2. Make a separate column for each premise and for the conclusion. Add intermediate columns as needed.
3. Mark the rows where *all* premise columns equal 1.
4. Inspect the conclusion *only on those rows.* A single 0 proves invalidity; otherwise the argument is valid.
5. State the verdict in a complete sentence, including the counterexample interpretation if invalid.

4 Rules of inference

A **rule of inference** is a known valid argument form. Rather than build a truth table for every new argument, we can apply these rules to premises and to previously derived statements.

4.1 Modus ponens (MP)

$$\frac{p \to q \qquad p}{q}$$

If p guarantees q and p is true, then q must be true. This is the alarm example.

4.2 Modus tollens (MT)

$$\frac{p \to q \qquad \sim q}{\sim p}$$

If p guarantees q, but q is false, then p cannot have been true. For example, if being a multiple of 8 implies being even, an odd integer cannot be a multiple of 8.

4.3 Hypothetical syllogism (HS)

$$\frac{p \to q \qquad q \to r}{p \to r}$$

A chain of guarantees can be shortened: p guarantees q, which guarantees r, so p guarantees r.

4.4 Disjunctive syllogism (DS)

$$\frac{p \vee q \qquad \sim p}{q}$$

At least one of p, q must hold. If p does not hold, then q must hold. The same works with the roles exchanged.

4.5 Conjunction, simplification, and addition

Rule	Valid form	Explanation
Conjunction	$p,\ q\ \therefore\ p \wedge q$	Combine two established truths.
Simplification	$p \wedge q\ \therefore\ p$	Either conjunct follows from a conjunction.
Addition	$p\ \therefore\ p \vee q$	If p is true, the disjunction is true regardless of q.

4.6 Resolution

$$\frac{p \vee q \qquad \sim p \vee r}{q \vee r}$$

If p is true, the second premise forces r; if p is false, the first forces q. Either way, $q \vee r$ holds.

Key idea. A rule may be applied only when the premises match its form. You cannot reverse a rule simply because the reversed statement looks plausible.

5 Formal derivations with numbered lines

A formal derivation is a list of statements. Every line must be either a premise or a consequence of earlier lines, justified by a named rule. The line numbers matter because they show *which* statements support each step.

Worked example: a two-step derivation

Premises: $p \to q$, $q \to r$, p. Show r.

Line	Statement	Justification
1	$p \to q$	Premise
2	$q \to r$	Premise
3	p	Premise
4	q	MP, lines 1 and 3
5	r	MP, lines 2 and 4

Line 4 is not a new assumption. It is established by applying MP to two earlier lines. Once established, it can be used just like a premise.

Worked example: using disjunctive syllogism and conjunction

Premises: $p \vee q$, $\sim p$, $q \to r$. Show $q \wedge r$.

Line	Statement	Justification
1	$p \vee q$	Premise
2	$\sim p$	Premise
3	$q \to r$	Premise
4	q	DS, lines 1 and 2
5	r	MP, lines 3 and 4
6	$q \wedge r$	Conjunction, lines 4 and 5

The conclusion is a conjunction, so we first prove each component separately and then combine them.

6 Inference rules versus equivalence laws

An **equivalence law** says two expressions have identical truth values under every interpretation. For instance,

$$p \to q \ \equiv \sim p \vee q.$$

This lets us replace an expression by an equivalent one. An **inference rule** says a new statement follows from statements already known to be true. For example, MP lets us derive q from $p \to q$ and p.

The distinction matters in written work: when you *rewrite* a statement, cite the equivalence law; when you *derive* a new statement from earlier lines, cite the inference rule and line numbers. A derivation can use both, provided each step is justified.

Worked example: rewrite first, then infer

Premises: $\sim p \vee q$ and p. Show q.

Line	Statement	Justification
1	$\sim p \vee q$	Premise
2	p	Premise
3	$p \to q$	Conditional equivalence, line 1
4	q	MP, lines 3 and 2

7 Common invalid argument forms

These errors are worth recognizing because they look superficially like legitimate inference rules.

7.1 Affirming the consequent

$$\frac{p \to q \qquad q}{p} \qquad \text{INVALID}$$

Counterexample: $p = 0, q = 1$. Both premises are true, but the conclusion is false. "If it rains, the ground gets wet; the ground is wet; therefore it rained" ignores other ways for the ground to become wet.

7.2 Denying the antecedent

$$\frac{p \to q \qquad \sim p}{\sim q} \qquad \text{INVALID}$$

Counterexample: $p = 0, q = 1$. The first premise and $\sim p$ are true, while $\sim q$ is false. A false hypothesis tells us nothing about whether the conclusion happens to hold.

7.3 Confusing $\vee$ with exclusive choice

In propositional logic, $p \vee q$ means *at least one* is true, including the case where both are true. From $p \vee q$ and p, we cannot conclude $\sim q$. That would require an additional premise excluding the possibility that both are true.

8 Choosing a method

Goal	Useful approach
Determine whether an argument is valid	Truth table; inspect critical rows
Disprove validity	Give one interpretation with all premises true and conclusion false
Prove a conclusion from given premises	Numbered derivation with named inference rules
Show two formulas mean the same thing	Truth table or equivalence-law derivation
Diagnose a suspicious argument	Compare with known valid and invalid forms

9 Common mistakes and grading pitfalls

1. **Checking every false conclusion.** Only a false conclusion on a *critical row* invalidates an argument.
2. **Calling a valid argument sound automatically.** Soundness additionally requires all premises to be true.
3. **Affirming the consequent.** $p \to q$ and q do not imply p.
4. **Denying the antecedent.** $p \to q$ and $\sim p$ do not imply $\sim q$.
5. **Skipping steps in a derivation.** State each intermediate conclusion on its own numbered line.
6. **Omitting rule names or line numbers.** A correct formula without its justification may not earn full credit.
7. **Using an equivalence law as though it were an inference rule.** Explain whether you are replacing an equivalent formula or deriving a consequence.
8. **Using a truth table when the problem requires a formal derivation.** Both are useful, but they are not interchangeable when a method is specified.

10 Practice problems

Attempt each problem before consulting the solutions. Use 1 and 0 for truth values, and descending binary row order.

1. Define a valid argument in terms of interpretations. What makes an argument invalid?

2. Use a truth table to test: $p \to q$, p; therefore q.

3. Use a truth table to test: $p \to q$, q; therefore p. Give a counterexample if invalid.

4. Determine whether $p \vee q$, $\sim p$; therefore q is valid. Name the rule.

5. Identify the rule: $p \to q$, $q \to r$; therefore $p \to r$.

6. Write a numbered derivation from $p \to q$, $q \to r$, p to r.

7. Write a numbered derivation from $p \vee q$, $\sim p$, $q \to r$ to r.

8. Test the argument $p \to q$, $\sim p$; therefore $\sim q$. Give a counterexample if invalid.

9. Explain why the truth of a conclusion by itself does not establish validity.

10. From $p \wedge q$ and $q \to r$, derive $p \wedge r$ using numbered lines.

11. From $p \to q$ and $\sim q$, derive $\sim p$. Explain why the rule is valid.

12. From $p \vee q$ and $\sim p \vee r$, derive $q \vee r$ and identify the rule.

13. From $\sim p \vee q$ and p, derive q, citing one equivalence and one inference rule.

14. Is an argument with no critical rows valid? Explain using the definition.

11 Detailed solutions

1. Valid means every interpretation making all premises true also makes the conclusion true. Invalid means at least one interpretation makes all premises true and the conclusion false.

2.

p	q	$p \to q$	p	q
1	1	1	1	1
1	0	0	1	0
0	1	1	0	1
0	0	1	0	0

Only the first row is critical; its conclusion is true. Valid.

3.

p	q	$p \to q$	q	p
1	1	1	1	1
1	0	0	0	1
0	1	1	1	0
0	0	1	0	0

The third row is critical with a false conclusion, so the argument is invalid.

4. Valid, by disjunctive syllogism (DS). If p is false and at least one of p, q is true, q must be true.

5. Hypothetical syllogism (HS).

6.

1.	$p \to q$	Premise
2.	$q \to r$	Premise
3.	p	Premise
4.	q	MP, 1 and 3
5.	r	MP, 2 and 4

7.

1.	$p \lor q$	Premise
2.	$\sim p$	Premise
3.	$q \to r$	Premise
4.	q	DS, 1 and 2
5.	r	MP, 3 and 4

8. Invalid: set $p = 0, q = 1$. Then $p \to q = 1$ and $\sim p = 1$, but $\sim q = 0$. This is denying the antecedent.

9. A conclusion might be true by coincidence even when the premises do not force it. Validity requires checking *every* interpretation in which the premises are all true.

10.

1.	$p \land q$	Premise
2.	$q \to r$	Premise
3.	p	Simplification, 1
4.	q	Simplification, 1
5.	r	MP, 2 and 4
6.	$p \land r$	Conjunction, 3 and 5

11. By modus tollens: $p \to q$ rules out $p = 1, q = 0$. Since $q = 0$, p cannot be 1; thus $\sim p$ follows.

12. Resolution. In the case $p = 1$, the second premise forces r; in the case $p = 0$, the first premise forces q. Thus $q \lor r$ holds.

13.

1.	$\sim p \vee q$	Premise
2.	p	Premise
3.	$p \rightarrow q$	Conditional equivalence, 1
4.	q	MP, 2 and 3

14. Yes. If there are no interpretations in which all premises are true, there cannot be an interpretation with all premises true and the conclusion false. The argument is therefore vacuously valid, though its premises cannot all be true simultaneously.